

Sparse Topological Pharmacophore Graphs for Interpretable Scaffold Hopping

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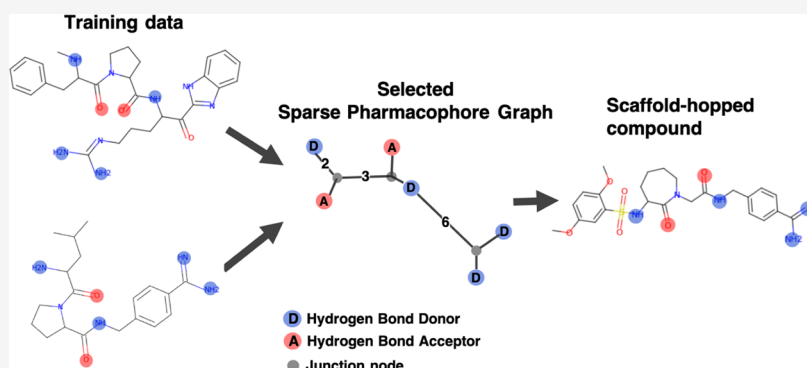
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ABSTRACT: The aim of scaffold hopping (SH) is to find compounds consisting of different scaffolds from those in already known active compounds, giving an opportunity for unexplored regions of chemical space. We previously demonstrated the usefulness of pharmacophore graphs (PhGs) for this purpose through proof-of-concept virtual screening experiments. PhGs consist of nodes and edges corresponding to pharmacophoric features (PFs) and their topological distances. Although PhGs were effective in SH, they are hard to interpret as they are complete graphs. Herein, we introduce an intuitive representation of a molecule, termed as sparse pharmacophore graphs (SPhG) by keeping the topological distances among PFs as much as possible while reducing the number of edges in the graphs. Several benchmark calculations quantitatively confirmed the sparseness of the graphs and the preservation of topological distances among pharmacophoric points. As proof-of-concept applications, virtual screening (VS) trials for SH were conducted using active and inactive compounds from ChEMBL and PubChem databases for three biological targets: thrombin, tyrosine kinase ABL1, and κ -opioid receptor. The performances of VS were comparable with using fully connected PhGs. Furthermore, highly ranked SPhGs were interpretable for the three biological targets, in particular for thrombin, for which selected SPhGs were in agreement with the structure-based interpretation.

INTRODUCTION

The aim of scaffold hopping (SH) is to find compounds consisting of different scaffolds from those in already known active compounds.^{1–4} Identifying scaffold-hopped compounds gives opportunities of improving bioactivities, different pathways of synthesis and circumventing other obstacles derived from absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties.^{1,2}

Ligand-based approaches for SH have been intensively investigated.^{5–13} As molecular representations for SH, two-dimensional topological pharmacophores,⁵ three-dimensional partial charge information,^{6–8} biological assay information,⁹ deep neural networks (NN) with graph attention mechanism,¹⁰ and multimodal transfer NN¹¹ were proposed. Furthermore, methods for generating scaffold-hopped chemical structures were also developed.^{12,13}

Among others, topological pharmacophores (TP) were successfully applied for identifying novel scaffold-hopped compounds in prospective applications,^{5,7,8} and related methods

have been developed.^{3,14–21} A pharmacophore is a geometrical relationship among parts of a molecular structure, which is responsible for biological activity or protein–ligand interaction. In a TP, the geometrical distances are replaced with topological distances to increase the degree of abstraction of the pharmacophore.

In this study, we investigate graph representations for TPs. Graphs consisting of pharmacophoric points are called pharmacophore graphs (PhGs). Nodes in a PhG are pharmacophoric features (PFs), and edges represent topological distances between PFs. Recently, Métivier et al. analyzed kinase

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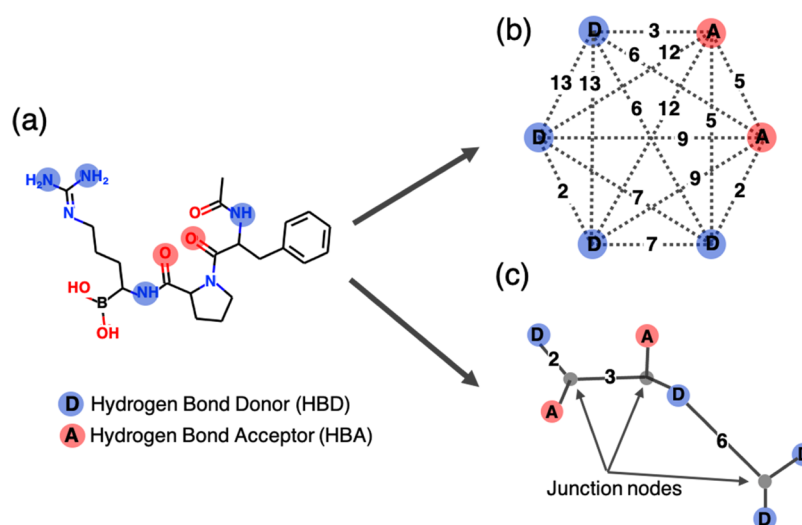


Figure 1. Conceptual difference between a pharmacophore graph (PhG) and a sparse PhG (SPhG). Difference between a PhG and an SPhG for an exemplary compound is illustrated. A combination of pharmacophoric features (PFs) is assigned on components (atoms, or sets of atoms) of the chemical structure (a). These PFs and their topological distances are converted into two graphs: a complete graph (PhG) (b) and a simpler graph (SPhG) (c). In the PhG, nodes correspond to PFs and edges to the topological distances between nodes (b). Similarly, the SPhG keeps topological information as much as possible, but has less edges by introducing junction nodes and eliminating redundant edges (c).

inhibitors to identify important PhGs based on the ratios of active and inactive compounds corresponding to the PhGs.²² They discussed hierarchical relationships of PhGs by network analysis and interpreted some of the PhGs based on corresponding molecular structures.

Employing PhGs as a set of queries for virtual screening focusing on SH was previously investigated by our group.²¹ While we can generate many PhGs from active compounds in the queries, only a part of PhGs reflects the important features for activity. Thus, the selection schemes made remarkable differences in SH performances and the proposed query ranking method *NScaffold* outperformed other conventional ranking methods, including simple counting of active compounds supported by a PhG.^{21,23,24} *NScaffold* ranks PhGs based on the number of their supporting scaffolds, not molecules. This trick prioritized PhGs which did not depend on single scaffold in retrospective validations.

PhGs themselves are hard to interpret because they are complete graphs: every pair of PFs connected by an edge.^{21,22} It is difficult to obtain insights from PhGs by visual inspection (*vide infra*) without corresponding chemical structures.

Molecular structures can also be recognized as graphs: nodes are atoms and edges are covalent bonds. And molecular graphs are naturally sparse because most types of atoms have at most a valence of four. Therefore, PhGs (complete graphs) could be reduced to simpler graphs to improve interpretability using the information of corresponding molecular graphs.

Reducing a molecular structure to an abstract form while keeping PFs was proposed to make “reduced graphs”. Rarey et al. introduced “feature trees (FTs)”, which are generated from molecules by replacing substructures with several types of nodes with respect to PF types and collapsing rings to a node.¹⁸ They also introduced efficient matching algorithms for the FTs.¹⁸ Barker et al. developed multilevel abstraction of molecular structures, which were converted to pairwise similarity or fingerprint for SH.¹⁹ Stiefl et al. proposed a different type of reduced graph, called “extended reduced graph”.²⁰ In an extended reduced graph, PFs are nodes and edges are either covalent bonds or virtual bonds connecting between a PF and an

atom (e.g., an atom connected with substitution in an aromatic ring). Although extended reduced graphs capture overall relations between PFs on chemical graphs, the algorithm does not capture precise topological paths among PFs when fused aromatic rings are involved. This makes precise graph matching difficult; however, this is important to extract common PFs among active compounds.

In this study, we introduce intuitive forms of PhGs, termed as “sparse pharmacophore graphs (SPhGs)” to keep the balance between interpretability and representation of precise topological distances among PFs. An SPhG is an abstract description of a PhG, and topological relationship (shortest path among PFs on a chemical graph) is maintained as much as possible. Newly introduced junction nodes make this representation possible. As our contributions, detailed algorithms for creating SPhGs from a chemical structure are described. Using around 900 000 compounds extracted from ChEMBL²⁵ and PubChem²⁶ databases, benchmark calculations of sparseness and distance-preserving ability of SPhGs are investigated. Furthermore, performances for SH in virtual screening tasks were compared between SPhGs and PhGs for three biological targets: Thrombin, Tyrosine Kinase ABL1, and κ -opioid receptor. Interpretations of highly ranked SPhGs are provided as a merit of using SPhGs.

Apart from a tiny difference of defining reduced graphs, such as ring reduction, the key difference between the existing reduced graph studies and ours is that (S)PhGs are introduced for extracting shared parts among active molecules: trying to be independent of individual molecules. In previous studies, a single reduced graph is extracted from a molecule for understanding the relationship between PFs occurring inside the molecule. Consequently, the algorithms of SPhG generation introduced in this study are optimized for this purpose and different from previous studies. Furthermore, we introduce aromatic bonds instead of aromatic ring features trying to represent more abstract features commonly existing among active molecules while keeping the topological distances in a molecule.

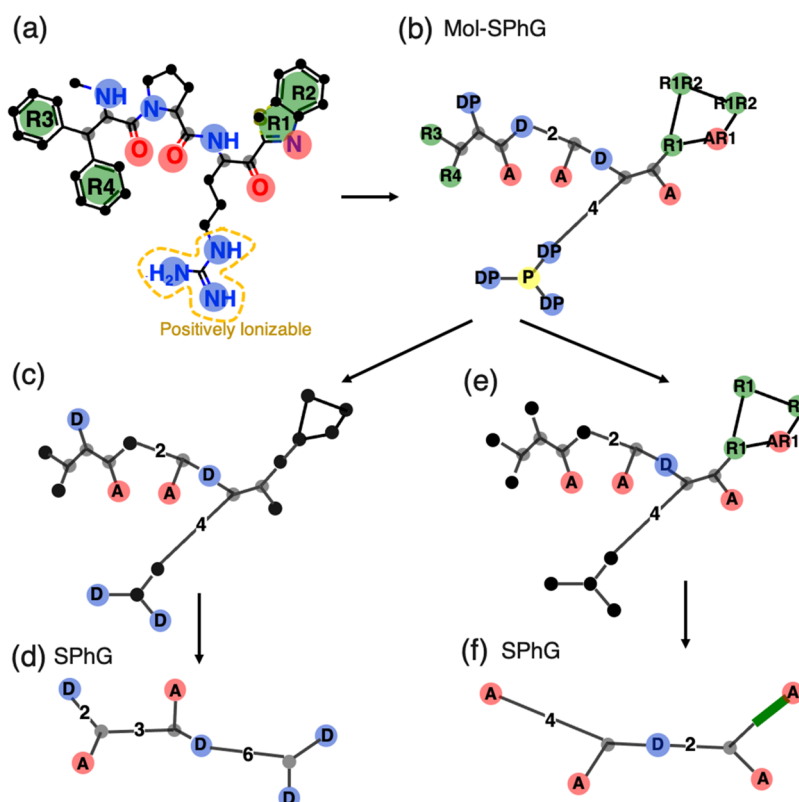


Figure 2. Generating SPhGs. Step-by-step graph reduction algorithms are applied to a molecule. (a) All PFs are assigned to components of the molecule structure. (b) The chemical structure with PFs is simplified to a Mol-SPhG by the graph reduction algorithm explained in Figure 3a. (c, d) From the Mol-SPhG, the exhaustive combinations of PFs with a predetermined number are selected and the unselected PFs turn to non-PF nodes. (e, f) The graph reduction algorithm in Figure 3b is then applied to the graphs, and the final SPhGs are created, accordingly.

MATERIALS AND METHODS

Sparse Pharmacophore Graphs (SPhGs). SPhGs are sparse forms of previously proposed pharmacophore graphs (PhGs).²¹ A PhG consists of nodes and edges, where nodes are atoms with pharmacophoric features (PFs), and edges represent shortest (topological) paths of two atoms with PFs. Therefore, a PhG is a complete graph: every node pair is connected by an edge. Figure 1a shows an exemplary chemical structure with a set of atoms with PFs. The extracted PhG is shown in Figure 1b. SPhGs, on the other hand, take two types of nodes: atoms with PFs and junction atoms. Junction atoms are introduced to preserve intra-node distances in the original chemical structure as much as possible after transforming to an SPhG. These atoms have more than or equal to three edges, and they are determined depending on a set of atoms with PFs (*vide infra*). Owing to the junction atoms, SPhGs become more intuitively understandable (Figure 1c) than the corresponding PhGs (Figure 1b).

Overview of Workflow for Extracting SPhGs. Figure 2 depicts stepwise modifications of an exemplary chemical structure to SPhGs by applying the proposed algorithms. First, all PFs are assigned to atoms. In Figure 2a, six atoms are annotated with HBD, four HBA, two are positive ionizables, and four points aromatic rings. All covalent bonds are converted to a single bond (edge), representing only connection between atoms. Second, the graph is transformed to a reduced graph called “Mol-SPhG” while keeping all of the PFs as nodes (Figure 2b). Detailed algorithms of transforming to Mol-SPhGs will be explained later in the manuscript. Third, predefined number of PFs are selected, and the remaining nodes turn non-PF nodes, as black circles in Figure 2c,d. Exhaustive combinations of PFs were

selected in this phase (only two of them are depicted in Figure 2). In the last step, another graph reduction algorithm is applied to each of the graphs derived by selecting a set of PFs (Figure 2e,f). Nodes with the PF of aromatic ring are converted to aromatic bonds, and only relevant bonds for the preservation of intradistance relationship are kept in the SPhG (Figure 2f).

In this study, the number of selected PFs was set to six to be consistent with our previous work and to avoid combinatorial explosion at the same time.²¹

Detailed Algorithms for Extracting SPhGs. Figure 3 shows the flowcharts of the proposed algorithms for generating SPhGs. The whole process can be divided into two parts: generating a Mol-SPhG from an input chemical structure (Figure 3a) and SPhGs from the Mol-SPhG (Figure 3b). A Mol-SPhG contains every atom annotated with PFs with a smaller number of nodes than in the original structure. It also keeps the shortest path between every pair of these annotated atoms. On the other hand, an SPhG contains a subset of these annotated atoms, but less edges and nodes than the Mol-SPhG while keeping the shortest-path relationship as much as possible. For making a Mol-SPhG, all PFs and junctions are assigned to atoms in a molecule structure. These processes are followed by the “graph reduction” process, which iteratively refine a graph to a simpler one by removing or merging non-PF or junction atoms (Figure 3a).

Graph Reduction Process. Figure 4 illustrates three operations in the “graph reduction” process to reduce nodes with non-PFs and with the PF of aromatic ring. One-edged nodes with non-PF are deleted because they are terminal nodes on the graph and do not contribute to the path-relation

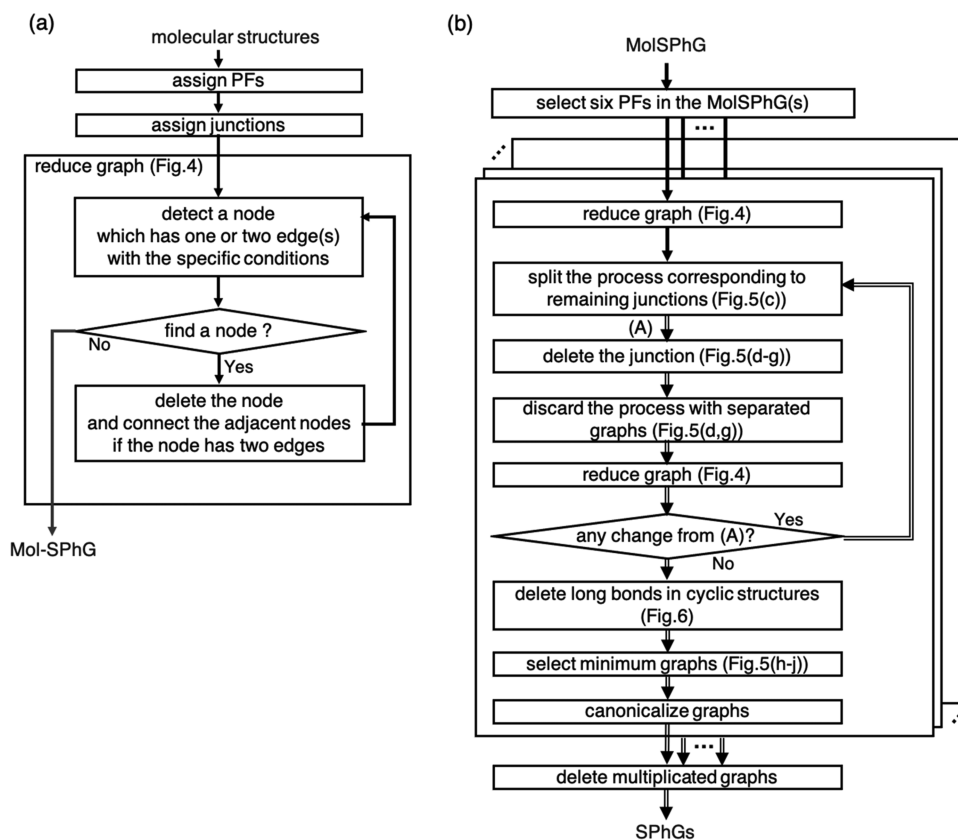


Figure 3. Flowcharts of the proposed algorithm for making SPhGs. (a) The procedure of converting a molecular structure to a Mol-SPhG. In this process, nodes with one or two edge(s) are iteratively discarded to find a reduced graph with the same distance relations as in the original molecular structure. The step denoted as “graph reduction”, described in Figure 4, deletes nodes and edges while keeping PFs as well as its topology. (b) The procedure of converting a Mol-SPhG to SPhGs with arbitrary selected six PFs. The unselected PFs are termed as non-PF nodes shown in Figure 2c,e. These non-PF nodes can be deleted by the “graph reduction” algorithm. SPhGs are made from the exhaustive six-node combinations of PFs. Arrows with double lines in (b) mean that more than one graph can be handled as a result of processing the step.

preservation (Figure 4a). Likewise, two-edged nodes are deleted by connecting the adjacent nodes directly. The distance between these adjacent nodes is preserved as the total topological distance between them before the deletion. In Figure 4b, the three two-edged nodes are deleted and the gray junction node and NH (PF node) are directly connected by a four-length edge. The terminal aromatic rings are truncated into a single node in the following way (Figure 4c). By applying the two-edged node deletion rule, explained above, two adjacent nodes emerged which are directly connected by two edges with different lengths (Figure 4d). Because two adjacent nodes are allowed to be connected by only one edge, the shortest-length edge is maintained (Figure 4e). The terminal node with a PF is merged to the adjacent node by applying the rule of deleting terminal node with non-PFs (Figure 4a). This rule can also be applied to adjacent nodes with the same PF having the same number. It should be noted that two adjacent atoms with the same type of PF are not merged, such as nitrogen atoms in an azo compound, because they are assigned to different numbers for the PF. In the exemplary case in Figure 4, the PF is the aromatic ring feature with number 3: “R3”. Therefore, one node with the PF having the same number survives in this case. These three operations are iterated until no eligible nodes for applying this rule are detected in the “graph reduction” process.

From Mol-SPhG to SPhGs. The second half of making SPhGs from a Mol-SPhG (Figure 3b) is described using an exemplary chemical structure in Figure 5. This process starts by selecting a

set of nodes with PFs from the Mol-SPhG in Figure 5b. Nonselected nodes with PFs turn non-PF nodes, and the “graph reduction” process is applied to the graph (Figure 5c). For further simplification of the graph, junction nodes are detected as shown in Figure 5c. In this example, four junction nodes are annotated with (d) to (g) in red. Then, each detected junction node is tested by deletion (Figure 5d–g), and only connected graphs are considered subsequently. Figure 5e,f shows single connected graphs, but the graphs in Figure 5d,g are disconnected. The latter ones are discarded, and the “graph reduction” process is applied to only the former graphs. If other junctions are detected in the newly created graphs, this testing process, i.e., deleting a junction and selecting only directed graphs, is iteratively applied until no change of the graphs occurs. To make SPhGs simpler and more intuitive, aromatic ring features are converted to edge-based description as shown in Figure 5h,i. Edges with two nodes sharing the same aromatic ring features are changed to bold and green lines without using the nodes of aromatic features. For example, in Figure 5h, the aromatic ring features of “R1” and “R2” are replaced with edge-based green bold lines, while feature “R3” remains as a node-based representation because of only one node having the feature. The next step is to delete long bonds in cyclic substructures (Figure 6). This process is a refinement procedure for fused rings (Figure 6a) and macrocyclic structures (Figure 6b). Here, edges whose lengths are equal to or longer than other path lengths for the adjacent atoms are iteratively deleted. In the

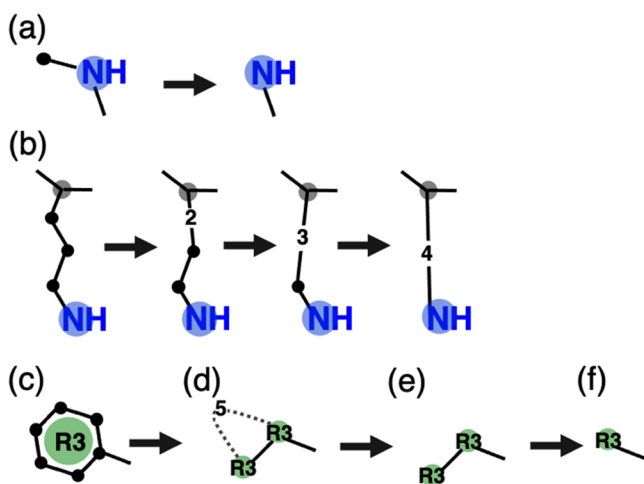


Figure 4. Graph reduction algorithm for single- and double-edge nodes. (a) Non-PF nodes are merged based on the following rules. One-edged non-PF nodes are deleted. (b) Non-PF nodes with two edges are replaced with an edge with the length of the sum of the two edges. (c–f) Terminal aromatic rings are reduced to a single node as follows. By repeatedly applying the above algorithm to the structure (c), two nodes with two edges having different lengths emerge, which violates the precondition of two nodes connected by a single bond. The shortest edge is prioritized, and the other edge is deleted (e). Further simplification leads to a single node (f) by applying the rule of removing the nodes other than connecting to different parts of the graph.

final step, a graph with the minimum total length of edges is selected (Figure S5). This process may output more than one graph when multiple graphs have the same minimum total length. After the selection, the graph(s) is canonicalized by means of the algorithm proposed by Schneider et al. and the

canonicalized graphs are called “SPhGs”.²⁷ Afterward, multiplied graphs among SPhGs for all of the combinations of six PFs are deleted.

Pharmacophoric Features (PFs). PFs are determined by the atomic environments of a focused site. Linear notations like SMARTS²⁸ and SLN²⁹ are usually used for identifying PFs. In the present study, employed PFs were hydrogen-bond donors (HBD), hydrogen-bond acceptors (HBA), aromatic rings, and positive and negative ionizables. A PF can be assigned to a set of multiple atoms. For example, an aromatic ring feature is usually assigned to five or six atoms, and in the guanidium fragment as shown in Figure 2a, a positive ionizable feature is assigned to three nitrogen and the center carbon atoms. As a definition of PFs, we employed RDKit implementation described under the file name of “BaseFeatures.fdef”.^{30,31}

Evaluation of SPhGs. Two important aspects of SPhG are preservation of topological distances and sparseness of the graphs. Previous researches including our study demonstrated that topological distances between PFs were important for controlling activity of drug candidates.^{3,14,21} Therefore, virtual screening (VS) of bioactive compounds using PubChem data sets was conducted using SPhGs as queries. The latter condition is required for enhancing the readability of (S)PhGs (Figure 1). Chemical structures are regarded as sparse graphs because most of the chemical elements in organic chemistry can take at most four bonds (edges). Sparseness also helps scientists in synthesis and medicinal chemists imagine possible molecular structures in modeling a graph. The sparseness of SPhGs and preservation of intradistance relationship were quantitatively measured, in addition to the discussion of interpretability.

Sparseness and Distance Preservation. Sparseness can be evaluated by sparse index defined as

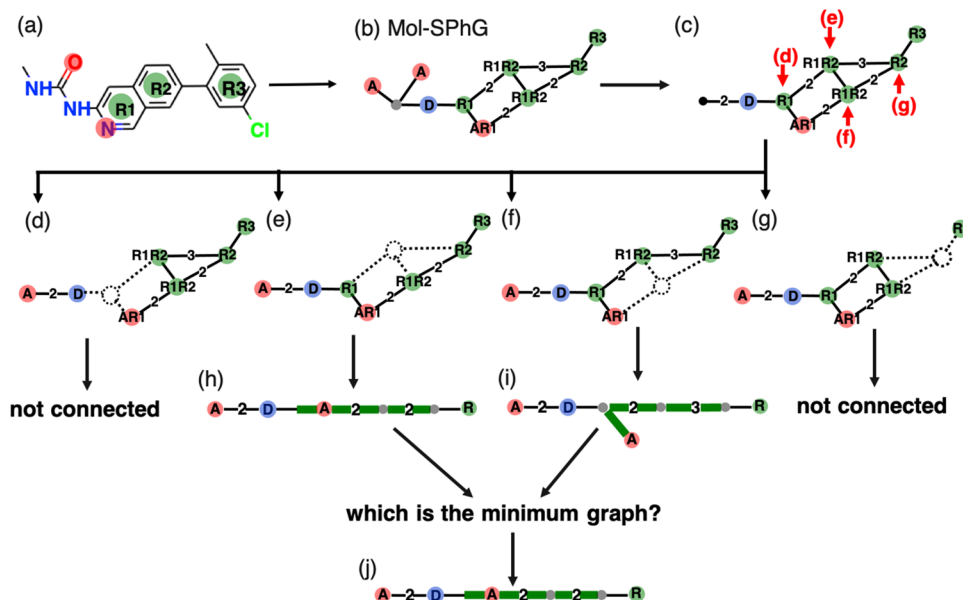


Figure 5. Conversion of ring structures to bonds. PFs are assigned to the active molecule for Tyrosine kinase ABL1 in (a). The molecule structure is converted to the Mol-SPhG as shown in (b). After generating Mol-SPhG with seven PFs (two donors, two acceptors, and three aromatic rings), six PFs except a donor on the leftmost position are selected and graph reduction method is applied to Mol-SPhG. As a result, we got the graph (c). Note that we have to generate all of the combinations of six PFs out of all PFs in Mol-SPhGs. The graph (c) is an example of all combinations. Four junction nodes are indicated by red arrows with labels (d–g). Each junction node is deleted and only connected graphs remain (e, f). The remaining graphs are converted to simplified ones (h, i). The graph(s) of the shortest total edge distance in (j) is(are) selected for processing further refinement.

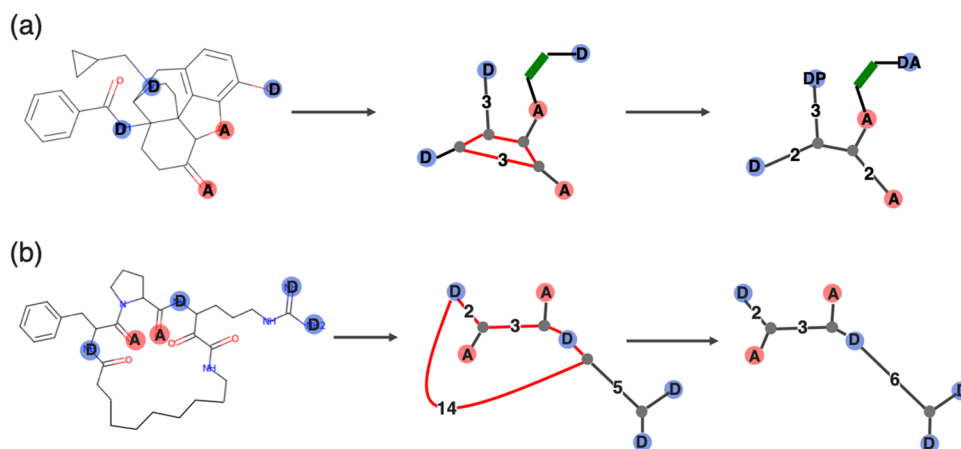


Figure 6. Removal of longer edges for SPhGs. Refinement procedure of graphs to SPhGs is applied to a fused-ring structure (a) and a macrocyclic structure (b). The edges whose lengths are equal to or longer than those of different paths are deleted.

Table 1. Compound Data Set Profiles^{abc}

ChEMBL ID ⁴	target	code	#CPDs ^d (highly potent CPDs ^e)	#Inactive CPDs for performance validations ^f	#Scaffolds		
					Bemis Murko	CCR Single	CCR RECAP
CHEMBL1862	tyrosine kinase ABL1	ABL1	601 (511)	346 990	200	97	254
CHEMBL204	thrombin	Thr.	1387 (514)	211 784	785	575	915
CHEMBL237	κ -opioid receptor	Kop.	2516 (1425)	284 546	1237	883	1506

^aBemis Murko: Bemis Murko scaffold using RDKit.^{30,32} ^bCCR Single: Compound core relationship (CCR)-based scaffolds.³³ ^cCCR RECAP: CCR-based scaffolds, but bonds were dissected based on the RECAP rules.^{33,34} ^dCPDs: all compounds for a target in the present study including active and inactive ones.²⁵ ^eHighly potent CPDs: compounds exhibiting pK_i values greater than or equal to 6.0. ^fInactive compounds filed in PubChem.²⁶

$$\text{sparse index (SPhG)} = \frac{N_E}{N_N - 1}$$

where N_E is the number of edges and N_N is the number of nodes. Tree structures are proved to have $N_N - 1$ edges; in other words, the sparse index of a tree is always 1. Ratios of SPhGs preserving the shortest distances between PFs are surveyed by calculating the distances identified with that of the corresponding PhG with the same selected PFs on the same molecules.

Virtual Screening Using SPhG. SPhGs can be used as queries for virtual screening (VS). Similar to our previous paper studying PhG, candidate compounds with at least one SPhG in the query set were regarded as hit compounds.²¹ The three scoring methods: (a) *Coverage*, (b) *NScaffold*, and (c) *Growth-Rate* were applied to prioritize SPhGs. A scoring method *Coverage*

$$\text{Coverage (SPhG)} = N_{\text{SPhG}}$$

where N_{SPhG} is the number of active compounds covered by the SPhG. Another scoring method *Growth-Rate*²² is the ratio of the *Coverage* score to the number of the inactive compounds covered by the SPhG defined as

$$\text{Growth-Rate (SPhG)} = \frac{\text{Coverage (SPhG)}}{N_{\text{inactive, SPhG}}}$$

where $N_{\text{inactive, SPhG}}$ is the number of inactive compounds covered by the PhG. In this study, compounds were assumed to be inactive when their pK_i values were less than 6.0. Here, the active and inactive compounds are taken from ChEMBL database.²⁵

The other scoring method *NScaffold* is the number of scaffolds covered by an SPhG, instead of the number of active compounds, i.e., *Coverage*. The definition of *NScaffold* is

$$\text{NScaffold (SPhG)} = N_{\text{scaffold, SPhG}}$$

Identifying SPhGs that are shared by different scaffolds is expected to be equivalent to the extraction of fundamental hypotheses for activity. Therefore, selected SPhGs based on *NScaffold* could identify new chemotypes, which are composed of scaffolds different from ones in the training active compounds.

Performance Measures for Scaffold Hopping. Performances of VS can be compared in the ability of identifying new chemotypes (scaffold-hopped compounds). Three types of scaffold definitions: Bemis Murcko (BM) scaffolds³² and two variants of scaffolds based on compound core relationships (CCR) were applied.³³ For deriving CCR-based scaffolds, covalent bonds in molecules were dissected when they were single (CCR Single) or defined by retrosynthesis rules (CCR RECAP).^{33,34} The data sets used in this study were extracted from the compounds whose activity information was registered in the ChEMBL database. To evaluate scaffold hopping performances, all compounds for each target were split into training and test data sets. In the same way as reported in our previous paper, all compounds in ChEMBL data sets are categorized into groups according to their scaffolds. Then, group-wise random split is conducted so that the ratio of training and test data sets became 1:4.²¹ The splits of the scaffold combinations were randomly chosen 10 times for each of the three scaffold definitions. A total of thirty trials for the three scaffold types were averaged in VS performances.

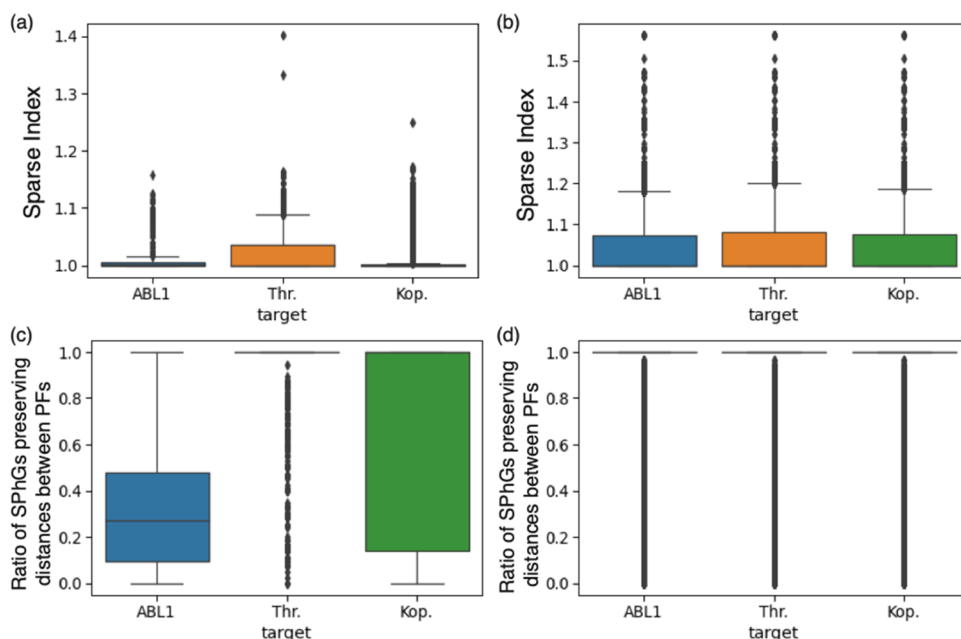


Figure 7. Statistics of SPhGs compared with PhGs. Boxplots of sparse index of the SPhGs of each compound in (a) ChEMBL and (b) PubChem inactive compounds. Boxplots of the ratios of SPhGs which conserve the shortest distances between PFs in each compound in (c) ChEMBL and (d) PubChem inactive compounds. In the boxplots, the boxes are defined as quartiles. The horizontal lines above the boxes indicate 1.5 times of widths between the medians and upper quartiles. In all of the three plots, the median is equal to the lowest values, i.e., one because more than half of the sparse indexes of SPhGs is one. There are no boxes in the boxplot due to most of the values being one except outliers.

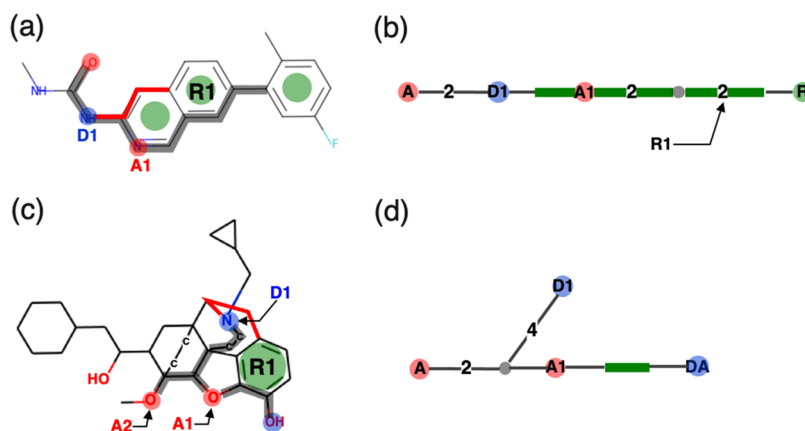


Figure 8. Examples of the SPhGs which do not conserve the shortest paths in the PhGs. (a) An example of the SPhGs in ABL1 which do not conserve the shortest paths between a PF and another. The bold black line shows a path for making SPhG via the process in Figure 3. On the contrary, the shortest paths between the aromatic ring R1 and the HBD D1 shown as a red path is different from the black path. The HBA A1 makes the bold black path circumvent the red one because the black one must pass through or reach the other PFs and keep the total lengths at minimum. (b) The corresponding SPhG to (a). (c) Another example in Kop. The shortest path between R1 and D1 (red path) and the paths for SPhG (the bold black path) is different.

A set of SPhGs are used as queries for VS. A test compound is predicted as active if at least one SPhG in the query set is found in the SPhGs pool from the test compound. In the current form of our algorithm, no fuzziness is allowed, meaning only equality between two SPhGs is evaluated.

Screening performances were evaluated using the values of correctly identified active compounds (TP), wrongly predicted active compounds (FN), and wrongly predicted inactive compounds (FP). Here, the inactive compounds in the test data sets were replaced with the compounds registered as inactive in PubChem database to increase the size of the data set.

Precision ($TP/(TP + FP)$) and recall ($TP/(TP + FN)$) were derived and plotted in the graphs in the Results and Discussion

section. The scaffold-based recalls and precisions were also calculated by replacing TP, FP, and FN with the scaffold-based numbers, which were calculated by the number of unique scaffolds in the compounds of TP, FP, and FN, respectively.

Compound Data Sets. As shown in Table 1, three macromolecular targets were selected from the ChEMBL database²⁵ version 24 based on the number of compounds listed for the targets, and macromolecular types: Tyrosine kinase ABL1 (ABL1), Thrombin (Thr.), and κ -opioid receptor (Kop.). Bioactive compounds with high confidence score were extracted and annotated with equilibrium constant (K_i) values. When multiple K_i values were available for a compound, their geometric mean was calculated to yield its final potency value

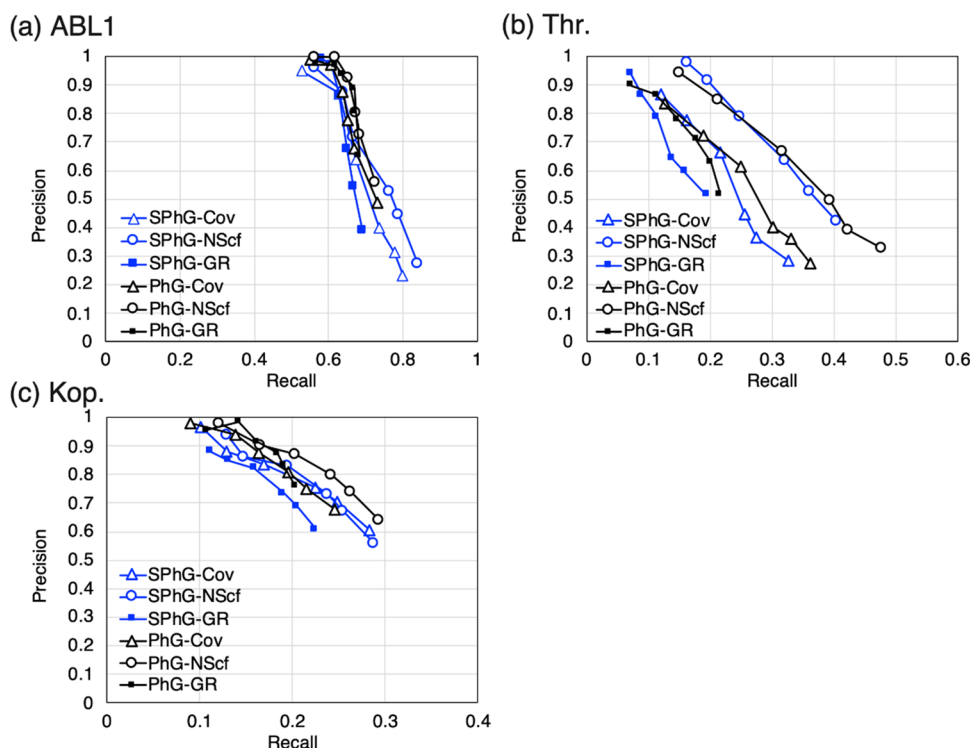


Figure 9. Virtual screening performance using PhG or SPhGs selected based on three different scoring methods. Target-wise comparison of PhGs and SPhGs in screening performance (training/test ratio = 1:4) for (a) Tyrosine kinase ABL1 (ABL1), (b) Thrombin (Thr.), and (c) κ -opioid receptor (Kop.). Reported are target-based screening performances in terms of averaged precision and recall as the number of PhGs or SPhGs in the query set was changed from 1 to 100. These performances are separately evaluated when three graph-selection algorithms (Coverage: Cov, NScaffold: NScf, and Growth-Rate: GR) are used as in the previous study.²¹ Screening was repeated 10 times by randomly splitting training and test data sets with a ratio of 1:4 on the basis of BM, CCR Single, or CCR RECAP scaffolds (10 trials each).

as long as all of the values fell into the same order of magnitude. In this study, compounds extracted from the ChEMBL database whose pK_i values were more than and equal to 6.0 were regarded as active. Compounds having pK_i values of less than 6.0 were also regarded as inactive along with inactive compounds in PubChem.

Data set profiles are reported in Table 1. Compounds with the molecular weight of less than 200 and more than 600 were discarded. For the three targets, the number of scaffolds ranged from 200 to 1237 in the BM scaffolds definition, 97 to 883 in CCR Single and 254 to 1506 in CCR RECAP. To calculate the precisions, 346 990, 211 784, and 284 546 inactive compounds for ABL1, Thr., and Kop. were used from PubChem, respectively.²⁶

RESULTS AND DISCUSSION

Study Design. PhGs have been shown to be effective for scaffold hopping—identifying isofunctional compounds with different scaffolds—using them as queries.²¹ However, PhGs are hard to interpret without corresponding chemical structures as they represent complete graphs (Figure 1). Although graphs with a small number of edges (sparse graphs) can be easily interpreted, transforming to a sparse form sometimes does not keep the shortest distances between pairs of nodes in a chemical structure. Our proposed SPhGs were designed to satisfy both the requirements as much as possible: preservation of shortest distances among atoms with PFs and interpretability of the graphs. In this study, the former feature was evaluated by simply counting the number of SPhGs preserving the shortest-path relationship, and the latter was qualitatively evaluated by visual

inspection. Furthermore, as a benchmark calculation, retrospective VS trials were conducted using a set of SPhGs as queries. The VS trials were intended to help us understand the effectiveness of SPhGs as queries. Some of the SPhGs lose the shortest-path relationship during the simplification of graphs. Thus, the performance of VS using the SPhGs was expected to be worse than using PhGs as queries.

Throughout benchmark calculations in this study, the number of PF nodes was fixed to six for compatibility with our previous study.²¹

For evaluating sparseness, SPhGs of active and inactive compounds for the three different biological targets: ABL1, Thr., and Kop., from ChEMBL and PubChem (Table 1) were used.^{25,26} The ratios of the SPhGs which preserved the shortest distances between PFs were also counted.

For evaluating the effectiveness of SPhGs being a set of queries for VS, the same data sets were used. Graphs (SPhG or PhG forms) of active compounds were used as a set of queries in VS trials. In the VS evaluations, the performance of identifying scaffold-hopped compounds was also calculated. Three different types of scaffolds: BM, CCR Single, and CCR RECAP were employed for this purpose. Both active and inactive compounds from ChEMBL were split into training and test data sets so that no scaffolds were shared in the two data sets. Furthermore, the ratio of the number of scaffolds in the training and test data sets was controlled as 1:4. This data split was randomly conducted 10 times for each of the three scaffolds. Precision and recall values were averaged for the whole 30 data sets to ensure statistical validness. For both types of VS trials: compound- and scaffold-based, query sizes were set from one (single graph) to

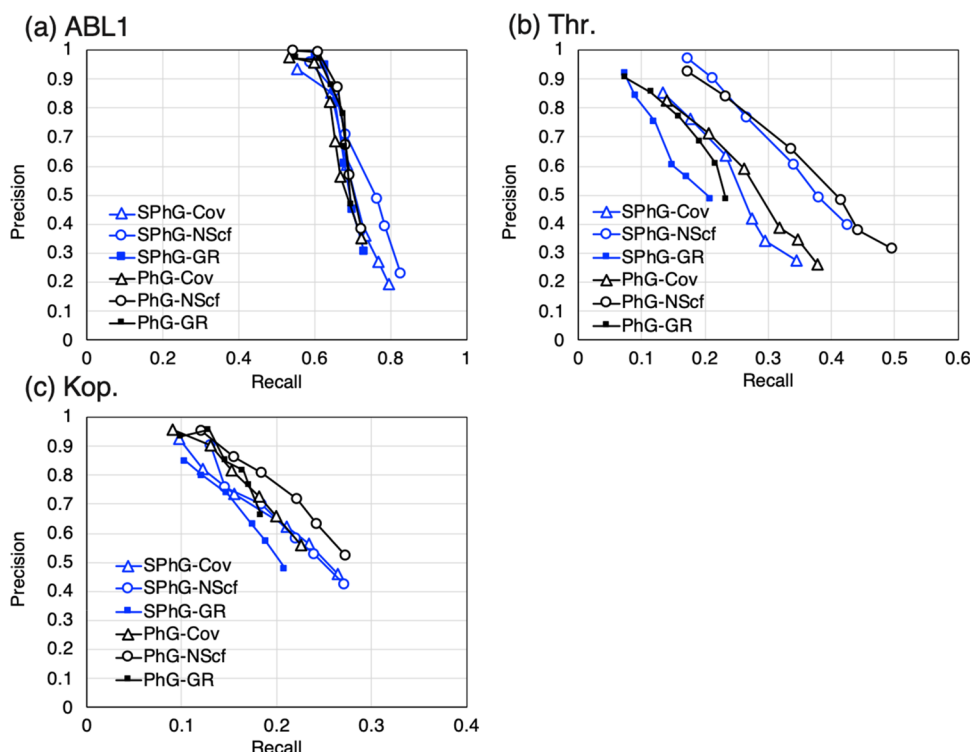


Figure 10. Virtual screening performance using PhG or SPhGs selected based on three different scoring methods (scaffold-based statistics). Reported are scaffold-based screening performances for (a) Tyrosine kinase ABL1 (ABL1), (b) Thrombin (Thr.), and (c) κ -opioid receptor (Kop.) in terms of averaged precision and recall as the number of PhGs or SPhGs in the query set was changed from 1 to 100. These performances are separately evaluated with three graph-selection algorithms (*Coverage*: Cov, *NScaffold*: NScf, and *Growth-Rate*: GR) also used in the previous study.²¹ The procedure was repeated 10 times for each scaffold type. Training and test data sets were randomly split with a ratio of 1:4 on the basis of BM, CCR Single, or CCR RECAP scaffolds (10 trials each).

100 for obtaining contentious metrics of VS performance. The query SPhGs (or PhGs) were selected based on the scores using the three criteria: *Coverage*, *NScaffold*, and *Growth-Rate*. For *Growth-Rate*, less potent active compounds were used as a set of inactive compounds in training data sets, for compatibility with a previous study.²¹ On the other hand, as a negative pool in test data sets, PubChem inactive compounds were used. The VS trials in this study evaluated the ability of finding active compounds of hundreds to a thousand from more than 200 thousand inactive ones.

Comparison of Sparseness and Preservation of Shortest Topological Distances in SPhGs. Sparse index values of SPhGs for the three targets were compared. For each compound, sparse index values were averaged over a set of SPhGs derived from the compound. The distributions of the averaged values are shown as boxplots. Figure 7a reports ChEMBL compounds and Figure 7b PubChem inactive. The sparse index values of PhGs were always three because a complete graph with six nodes has 15 edges. On the other hand, for active compounds, those of SPhGs ranged from 1.0 to 1.2 with few exceptions (Figure 7a). This means that SPhGs were tree-like structures (graphs without closed-loops). The same trend was observed when using inactive compounds from PubChem for the three targets (Figure 7b). In all of the boxplots in Figure 7a,b, the median values were exactly one because more than half SPhGs were tree structures (53.2% in ABL1, 50.0% in Thr., and 52.3% in Kop.).

Distributions of the ratios of SPhGs keeping the shortest-path relationship of the original PFs are reported in Figure 7c,d for ChEMBL and PubChem inactive, respectively. Each dot in these

plots represents a compound, and for each compound, ratio was calculated by dividing the number of distance-preserving SPhGs by the number of all SPhGs. Average ratios of the number of distance-preserving compounds in ChEMBL were 32.6, 91.7, and 64.6% for ABL1, Thr., and Kop., respectively. This indicates that it was difficult for active compounds (including weakly potent compounds) against ABL1 and Kop., to maintain topological distances between nodes with PFs once they were converted to SPhGs.

However, for PubChem inactive compounds, there was no significant difference in the number of compounds preserving the distances among targets as shown in Figure 7d. Actually, 79.3% of compounds in ABL1, 79.2% in Thr., and 79.2% in Kop. maintained the shortest distances between PFs.

Figure 8 exemplifies two SPhGs which did not preserve the shortest-distance relationship in PhGs. Figure 8a shows an active compound for ABL1 with a set of selected PFs. The shortest path length from the aromatic ring “R1” to the HBD “D1” is three, as drawn in red bold line. However, in the corresponding SPhG (Figure 8b), a different path was selected, as drawn in gray bold line, resulting in the path length of four between the two PFs. This was because the existence of another PF: “A1” does not allow to select the red path for making a graph with the minimum number of edges connecting all of the PFs. The other nondistance preservation example is for Kop., as shown in Figure 8c,d. The shortest path from “R1” to “D1” is shown in red bold line (path length was three), while the corresponding SPhG goes along the gray line due to the existence of another PF: “A1”. Overall, distance preservation in SPhGs depended on the combinations of selected PFs. If we chose three PFs: A1, D1, and

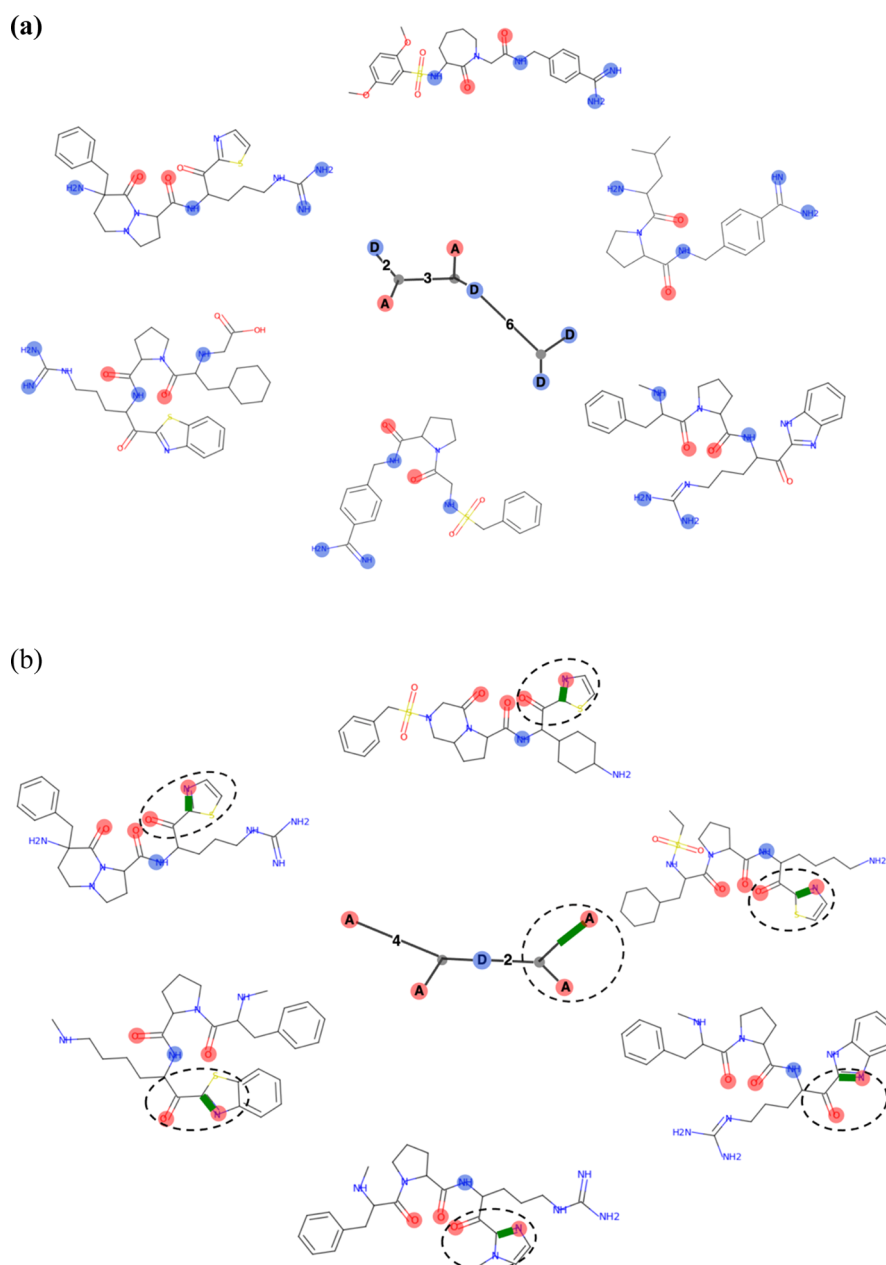


Figure 11. Exemplary active compounds sharing the same SPhGs (thrombin). For each SPhG, six compounds containing the SPhG are displayed. Green lines in (b) means aromatic bond features.

A2 in Figure 8c, a created SPhG would keep the shortest-distance relationship among them.

In summary, these analyses indicate that the conversion algorithm for making SPhGs succeeded to produce simpler tree-like structures while keeping distance relationship for most of the input chemical structures. Also, they clarified types of chemical structures for which distance preservation was difficult when transforming to SPhGs.

Comparison of Scaffold Hopping Performances.

Precision–recall curves for the three scoring methods by changing the number of SPhGs and PhGs as set queries ($N = 1–100$) are shown in Figure 9. The marks (squares, circles, and diamonds) of the lowest recall correspond to a single query $N = 1$ with recall values increasing as N increases. From left to right, six marks on each line correspond to $N = 1, 3, 10, 30, 50$, and 100. Performances using SPhGs are drawn in blue, while using

PhG in black. Most of the performances of SPhGs were comparable to those of PhGs. In this study, PhGs include additional two types of PFs: hydrophobic and zinc binders in the same way as in the previous study.²¹

In Figure 9a, the performances for ABL1 overlapped between SPhGs and PhGs at $N = 1, 3, 10$, and 30 while the precisions deteriorated at $N = 50$, and 100 by about 20%. For Thr., as shown in Figure 9b, the precision values when using the SPhGs, selected based on the *NScaffold* metric (“SPhG-NScf”) outperformed those using the PhGs (“PhG-NScf”) at $N = 1$ or 3, while the superiorities were flipped at $N = 10, 30, 50$, and 100. For Kop., as shown in Figure 9c, the differences in precision between SPhG and PhG were less than 10%. These performance gaps between SPhGs and PhGs for ABL1 and Kop. can be attributed to the small ratios of preserving the shortest distances between PFs, as shown in Figure 7c.

Figure 10 shows precision–recall curves using scaffold-based data sets. Figure 10a,b shows no remarkable differences from using the compound-based splitting shown in Figure 9a,b, respectively. In the case of Kop., although there were no differences in performance when using the *Coverage* method, PhGs showed better performance than SPhGs using *NScaffold*-based query selection. These results indicated that SPhGs showed slightly inferior scaffold hopping capability compared to PhGs. Figure S1g–i reports the recall–precision curves based on types of scaffolds for Kop. When using BM (Figure S1g) and CCR RECAP (Figure S1i) scaffolds, PhGs looked significantly better than SPhGs. Most of the highly ranked PhGs using these scaffold types contained a five-membered ring adjacent to an aromatic ring and a sterically bulky ring, as shown in Figure S3. On the other hand, when using CCR Single (Figure S1h), SPhGs were better than PhGs. In this case, some highly ranked PhGs had different characteristics (combination of PF nodes), which did not match any of the six molecules in Figure S3. This indicates that SPhGs have a limitation expressing these condensed reduced ring structures with multiple paths from a PF to another one. As discussed before, since distance preservation and sparseness of graphs showed trade-off relationship, using complex fused-ring structures in query graphs might be necessary for keeping high VS performance while using PhGs.

Identified SPhGs. Typical examples of the first- and second-ranked SPhGs based on the *NScaffold* criterion for Thr. are shown in Figure 11. The SPhGs shown in Figure 11a have four HBD and two HBA. Compared to the complete graph of PhG in the previous paper, as shown in Figure 1b, the SPhG is easy to interpret for the topological relations among the PFs (Figure 11a). Despite the sparsity, the essential features of interaction with Thr. were preserved. Like the interpretation proposed in the previous paper,²¹ all six PFs were consistent with the experimentally studied binding mode of thrombin inhibitors. The two HBDs sharing a junction with distant one corresponded to an arginine residue. This substructure was known to make an ionic interaction to the carboxyl group of Asp189.^{35–39} Furthermore, a pair of an HBD and HBA pair (amide group) with distance two is proven to bind with Ser214.³⁵ The other HBD and HBA pair sharing a junction with distance three were previously reported to interact with Gly216.³⁶ These binding sites were reported with the bound conformations measured by X-ray crystallography and registered in Protein Data Bank (PDB),³⁷ such as 3UTU,³⁶ 1C4V,³⁸ and 1LHC.³⁹

Figure 11b illustrates a typical second best SPhG for Thr. when using the *NScaffold* method. As indicated with black dashed circles, the SPhG has an aromatic bond feature with an HBA at the end and another HBA at the opposite side of the aromatic bond feature sharing a junction. These patterns do not exist in Figure 11a. The surrounding six molecules help interpret the SPhG in Figure 11b. Through the investigation of PDB,³⁷ we found these types of molecules bound to Thr., e.g., 1DOJ and 1A61.^{40,41} Ligands in both the complexes had the carbonyl part interacting with Gly193 and a nitrogen atom in an aromatic ring interacting with the residue of Hys57.^{40,41}

Likewise, for the other two targets, typical top-ranked SPhGs were shown along with typical active compounds in Figures S2 and S3 in the Supporting Information.

CONCLUSIONS

Herein, we have proposed sparse pharmacophore graphs (SPhGs) for scaffold hopping (SH). An SPhG is an abstract of the molecular structure of an active compound, focusing on pharmacophoric features (PFs). Nodes in an SPhG are PFs, and edges represent topological distances between PFs. Unlike previously proposed pharmacophore graphs (PhGs), interpretability of the graphs drastically improved owing to the sparseness of these graphs. Introduction of junction nodes, atoms not PFs, helped in keeping the topological distances among PFs while reducing the number of nodes. Detailed algorithms for compiling SPhGs are also provided, and several benchmark calculations quantitatively confirmed the sparseness of the graphs and the preservation of topological distances among pharmacophoric points.

As proof-of-concept applications, virtual screening (VS) trials for SH were conducted using active and inactive compounds from ChEMBL and PubChem databases, respectively, for three biological targets: thrombin, tyrosine kinase ABL1, and κ -opioid receptor. The performance of VS was comparable with using fully connected PhGs. And highly ranked SPhGs were interpretable for the three biological targets, in particular for thrombin, for which selected SPhGs were in agreement with structure-based interpretation.

We hope that the SH method proposed in this study becomes useful for medicinal chemists to try new chemotypes and modify candidate compounds based on topological pharmacophores.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c00409>.

Input files (ZIP)

Output files for boxplot (ZIP)

Output files (ZIP)

Detailed SH performances, additional SPhGs for ABL1 and Kop., and input/output data for evaluating SH performances, sparseness indices, and ratio of SPhGs preserving distances between PFs (PDF)

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Notes

The authors declare no competing financial interest.

The data supporting the findings of this study are available in the Supporting Information and the web-enhanced objects. Processes of reproducing the statistics shown in Figure 7 and the virtual screening results in Figures 9, 10, and S1 are given in Section S4 in the Supporting Information with descriptions of all files in the web-enhanced objects. The software developed for this study is available from the corresponding author upon request.

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